

Steven Kolawole

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RESEARCH FOCUS

The efficiency of LLM inference: model compression, KV cache optimization, adaptive serving (cascading and speculative decoding), and inference parallelism for production systems.

EDUCATION

Ph.D. in Computer Science, Carnegie Mellon University 2023 – 2028 (exp.)

Language Technologies Institute, School of Computer Science. Advisor: Prof. Virginia Smith.

B.Sc. in Computer Science, Federal University of Agriculture, Abeokuta 2017 – 2023

RESEARCH EXPERIENCE

Applied Scientist Intern, AWS AGI Foundations May – Aug 2026

Inference Optimization Science Team (Bedrock), with Nathan Pemberton and Kyle Ulrich. In-place speculative decoding via gated LoRA adapters, mask priming, and verifier-gated loss in quadratic and draft-then-verify topologies.

Graduate Student Researcher, Carnegie Mellon University Aug 2023 – Present

Advised by Prof. Virginia Smith. Algorithmic methods for efficient ML inference.

Independent Researcher, ML Collective Apr 2021 – Present

Pre-Ph.D. research mentored by Rosanne Liu and Jason Yosinski; now mentoring junior researchers within the same open-science pipeline. Resource-constrained NLP, federated learning, and efficient ML.

PUBLICATIONS

* denotes equal contribution.

- **S. Kolawole**, K. Santhanam, V. Smith, P. Thaker. *ParallelPrompt: Extracting Parallelism from Large Language Model Queries*. NeurIPS 2025, Datasets & Benchmarks Track. [OpenReview](#) · [arXiv:2506.18728](#).
- D. Soiffer, **S. Kolawole**, V. Smith. *Semantic Agreement Enables Efficient Open-Ended LLM Cascades*. EMNLP 2025, Industry Track. [ACL Anthology](#) · [arXiv:2509.21837](#).
- **S. Kolawole***, D. Dennis*, A. Talwalkar, V. Smith. *Agreement-Based Cascading for Efficient Inference*. Transactions on Machine Learning Research (TMLR), 2025. [OpenReview](#) · [arXiv:2407.02348](#).
- **S. Kolawole***, L. Dery*, J.-F. Kagy, V. Smith, G. Neubig, A. Talwalkar. *Everybody Prune Now: Structured Pruning of LLMs with only Forward Passes (Bonsai)*. [arXiv:2402.05406](#).
- **S. Kolawole**, O. Osakuade, N. Saxena, B. K. Olorisade. *Sign-to-Speech Model for Sign Language Understanding: A Case Study of Nigerian Sign Language*. IJCAI 2022, AI for Social Good Track. Proceedings · [arXiv:2111.00995](#).

MANUSCRIPTS UNDER REVIEW

- **S. Kolawole**, V. Smith. *Epiphany-Aware KV Cache Eviction Without the Attention Matrix (EpiKV)*. Under review, ICLR 2027. [arXiv:2606.26472](#).

- P. Jim*, **S. Kolawole***, O. M. Busoye, G. Bagai, V. Smith. *What Survives When You Compress a Recursive Reasoner for the Edge?* (EdgeTRM). Under review, ICLR 2027. [arXiv:2606.26488](#).

WORKSHOP PAPERS

- M. Oduwole, . . . , A. Owodunni, **S. Kolawole**. *From Scarcity to Efficiency: Investigating the Effects of Data Augmentation on African Machine Translation*. LM4UC @ AAAI 2026 (Springer CCIS). [OpenReview](#). [senior/last author]
- N. Obiefuna, S. Oyeneye*, S. Odunaiya*, I. Oyelaja*, **S. Kolawole**. *PrivacyBench: Privacy Isn't Free in Hybrid Privacy-Preserving Vision Systems*. ES-FoMo @ ICML 2025. [OpenReview](#). [senior/last author]
- B. Awobade*, M. Oduwole*, **S. Kolawole**. *What Happens When Small Is Made Smaller? Exploring the Impact of Compression on Small Data Pretrained Language Models*. AfricaNLP @ ICLR 2024. [arXiv:2404.04759](#). [senior/last author]
- C. Leong, . . . , **S. Kolawole**, et al. *Adapting to the Low-Resource Double-Bind: Investigating Low-Compute Methods on Low-Resource African Languages*. AfricaNLP @ ICLR 2023. [arXiv:2303.16985](#).
- N. Alam*, **S. Kolawole***, S. Sethi*, N. Bansali, K. Nguyen. *Vision Transformers for Mobile Applications: A Short Survey*. [arXiv](#), 2023. [arXiv:2305.19365](#).

PEER REVIEW & PROFESSIONAL SERVICE

- **ML/AI**: NeurIPS (2024, 2026); ICML (2026); ICLR (2025); AISTATS (2025, 2026); AAAI (2026); ES-FoMo @ ICML (2024).
- **NLP**: COLM (2025, 2026); ACL Demo (2025, 2026); EMNLP Demo (2024, 2025, 2026).

HONORS & AWARDS

- **Scholar, Machine Learning Summer School (MLSS) NYC** (Columbia University): competitively selected among ~200 Ph.D. researchers worldwide; fully funded. 2026
- **Research Grant, Algorand Foundation**: \$114,635 Ecosystem grant for an ML research-engineering project (the ASAnalytics platform). 2022
- **National Artificial Intelligence Champion**, Nigeria Computer Society: first place, national AI Challenge. 2022
- **Ideathon Winner** (\$5,000 Google Cloud credits), Deep Learning Indaba, the continent's foremost ML gathering. 2022
- **Best Poster, Data Science Nigeria**. 2021
- **Winner, DeepQuest AI Challenge**. 2021
- **Scholar, MTN Foundation Science & Technology Scholarship**: selection rate 1.8%. 2020, 2021
- **Led AI+ Abeokuta to Data Science Nigeria's "AI City of the Year"**: as community lead, produced the largest national-bootcamp cohort of any Nigerian city. 2020
- **Runner-up, "Mr. Algorithm," Data Science Nigeria**: nationwide community vote, for technical excellence and knowledge-sharing. 2020
- **Fellow, Young Professionals Bootcamp**: selection rate 3.6%. 2020
- **Gold Medal, NaijaHacks**: AgroAI plant-disease-detection app; top 12 of 336 teams. 2019

LEADERSHIP & COMMUNITY SERVICE

- **Founder & Lead, ML Collective Africa** 2023 – Present
Founded and lead the Africa program of ML Collective, a 501(c)(3) research non-profit, and act as its research lead: agenda, weekly research meetings, study programs, project advising, and a team of

organizers and mentors. Administer ML Collective’s Google Cloud compute grant (proposal review and allocation). Raised over \$50,000 since 2022 for African researchers’ Deep Learning Indaba attendance; chaired the program’s workshop at the Indaba 2026 (“Whose Intelligence? Building Agentic AI in Africa without Frontier Model Dependency”).

- **Campus Lead, AI+ FUNAAB; lead, AI+ Abeokuta** (Data Science Nigeria) 2019 – 2021
Built a structured remote preparation program that took the community from five national qualifiers (2019) to 38 of 42 entrants selected (2020; 28 in the national top 250, nine in the national top 20).
- **Community Lead, Google Developer Student Clubs, FUNAAB** 2021 – 2022
Technical training reaching 3,000+ students; 600+ taught personally.
- **Mentor, STEM for Development** 2023 – 2025
Mentor 20+ graduate-school applicants per year toward U.S. and other leading research programs.
- **Mentor, She Code Africa** 2021 – 2022
Mentored women entering machine learning and data science.

SELECTED TEACHING

Guest Lecturer (Model Compression) & Teaching Assistant, 10-718 Machine Learning in Practice, CMU Spring 2026

Invited Lecturer, AI Saturdays Lagos (AI6) Fall 2023

SELECTED INVITED TALKS & PRESENTATIONS

(Research orals marked †; invited talks/panels noted.)

- **2026:** “Doing ML Research from the Margins,” skills session, Deep Learning Indaba (Lagos); “A SOTA Overview of Efficient AI, and What It Means for Africa,” IndabaX Nigeria (*invited*).
- **2025:** Black in AI ELAI graduate-school talk (*invited*); CMU Africa Research Club (*invited*, ×2).
- **2024:** Low-resource LLM practicals, Deep Learning Indaba (Senegal).
- **2023:** Cohere For AI (C4AI) Independent-Research panel (*invited*); OSCAFEST.
- **2022:** IJCAI†; Deep Learning Indaba ML-efficiency talk (Tunisia); Lanfrica Talks (*invited*); EuroPython (Dublin); PyConDE & PyData Berlin; Berlin Buzzwords; Build Stuff; SciPy (Austin, Texas); NCS AI Summit; DataFest Africa; GDG Abeokuta DevFest (*invited*).
- **2021:** NeurIPS ML4D†; Masakhane Speaker Series (*invited*); PyData Global; EuroPython; PyCon APAC; Conf42 Python; ML Collective Research Jam (×3).
- **2020:** PyData Global; PyCon Africa.

SELECTED INDUSTRY EXPERIENCE

Founding ML Engineer, ASAnalytics 2022

Backend systems for real-time opinion-mining/analytics of blockchain’s digital assets; the vehicle for the Algorand Foundation research-engineering grant.

Data Science Intern, SeqHub Analytics LLC, New Haven, CT 2021

Fine-tuned mT5 for low-resource Nigerian-language machine translation; built microservices and a dockerized analytics dashboard.

SELECTED MEDIA

Interview on the sign-to-speech research, [AIhub](#) (2022); talk recordings on [PyVideo](#). Updated September 2026